

Exam Scoring Methodology - Montana RASS

Required under ARM 42.13.906(4)(d). One document covering both tracks.

Configuration

On-Premises Track (mt-alcohol)

- **Passing score:** 80%
- **Questions per attempt:** 40, drawn from a 64-question pool
- **Format:** Multiple-choice only
- **Time limit:** None (typical completion 15-20 minutes)
- **Retake policy:** Per ARM 42.13.907 as adopted March 7, 2026. If first attempt scores 70% or higher (but below 80%), the learner may retake the final exam only. If first attempt scores below 70%, the learner must retake the entire program before another final-exam attempt.
- **Score required:** 32 of 40 correct (80%)

Off-Premises Track (mt-alcohol-offsale)

- **Passing score:** 80%
- **Questions per attempt:** 25, drawn from a 40-question pool
- **Format:** Multiple-choice only
- **Time limit:** None (typical completion 10-15 minutes)
- **Retake policy:** Same ARM 42.13.907 rule as on-premises. First attempt 70-79%: retake the final only. First attempt below 70%: retake the entire program.
- **Score required:** 20 of 25 correct (80%)

Score calculation

Each question is worth 1 point. The score is the percentage of correct answers, calculated as:

$$\text{score_percent} = (\text{correct_answers} / \text{total_questions}) \times 100$$

A student passes when `score_percent >= 80`.

Question selection

Each attempt draws questions from the pool using a code-enforced minimum-per-outcome rule (specified in `final-exam.json#selectionRules.minimumPerOutcome` and applied by the exam player's question-selection function before any random fill):

- **Minimum number of questions per Knowledge Outcome** (K1, K2, ... Kn): every required content area is sampled every attempt.
- **Minimum number of questions per Performance Outcome** (P1, P2, ... Pn): every required practical skill is sampled every attempt.
- **Random selection within those minimums**: any two attempts draw a different set of 40 (or 25) questions.

Every module's content is mapped to module-specific Knowledge and Performance outcomes (see the module-to-outcome map in `curriculum-outline.md`), and every one of those outcomes carries a per-attempt minimum. As a direct consequence of the per-outcome rule, every module is represented in every attempt. This guarantees each attempt covers every statutory content area in ARM 42.13.906/.907.

Identity verification

Both tracks ship unproctored at launch, consistent with the Montana market norm (no approved provider currently uses live online proctoring at the general RASS level). Single-user account integrity is enforced via:

- Authenticated login with email + password (Supabase Auth)
- Captcha at signup and login
- IP rate limiting on signup (3/hour) and login (10/hour)

CertReady's AutoProctor SDK scaffold is wired in the platform code (`courses.is_proctored` per-course flag, `enrollments.proctoring_override` per-enrollment), so an employer can opt their employees into proctoring as a B2B perk on a per-enrollment basis. This is at the licensee's option, not a CARD requirement.

Retake handling — per ARM 42.13.907

The course enforces the adopted ARM 42.13.907 retake rule (effective March 7, 2026 per MAR Notice 2025-292.3 (adopting the amended proposal in MAR Notice 2025-292.2)):

1. **Score 80% or higher:** Pass. Certificate issued immediately.

2. **Score 70% – 79.99%:** Did not pass, but eligible to retake the final exam directly. Each new attempt draws a fresh random set of questions from the pool. No fee.
3. **Score below 70%:** Did not pass. Learner must retake the entire training program (all 8 on-premises modules / all 6 off-premises modules) before another final-exam attempt is permitted. The module-completion gates reset; the time-floor for the track must be re-met.

Every attempt is logged in `exam_sessions` (immutable audit trail) so any pattern of attempt-to-pass behavior is visible to CARD auditors on request.

A student who passes is issued a certificate immediately. The certificate is the final credential; passing students do not re-take the exam.

Treatment of borderline scores

A student scoring exactly 80% passes. A student scoring 79.99% (e.g., 19 of 25 = 76% on the off-premises track, between the 70% retake threshold and the 80% pass threshold) does not pass but is eligible to retake the final exam only. A student scoring 69.99% (or any score below 70%) does not pass AND must retake the entire program. The multiple-choice format is unambiguously machine-graded; no automatic re-grading or appeal.

Audit trail

Every exam attempt is logged with:

- `attempt_number`
- `score` and `max_score`
- `passed` (boolean)
- `retake_eligibility` ("final_exam_only" if 70-79.99%; "full_program" if below 70%; "passed" if 80%+)
- `answers_json` (full per-question record of student answer + correctness)
- `started_at` and `completed_at` timestamps
- `grading_method` ("multiple-choice")

Records are retained for 5 years per CertReady's record retention policy, exceeding the typical CARD audit horizon.